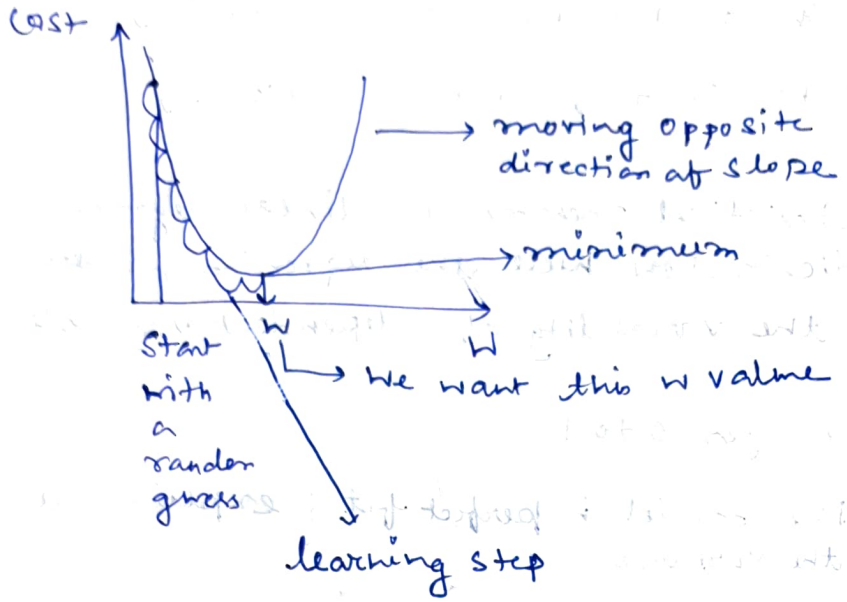
## Iterative optimization

- Start with a random guess for W.
- iteratively update it to minimize cost function.




## Gradient descent approach for iterative optimization

GD is an optimization algorithm we to minize a function by iteratively moving towards the direction of steepest descent (the negative gradient)

Here, we use for minimizing J(w)

- Imagine you are standing on a mountain and want to reach bottom.
- At each step, look the slope (gradient) and move downhill.
- If step is too big, might overshoot.
- If step is too small, take too long.

Partial derivatives give us the slope (i.e direction to move) with respect to a parameter.

## Multiple linear regression

$$y = XW + \epsilon$$

$m$ = no. of training samples
$n$ = no. of features.

Model: $\hat{y} = XW$

$$X_i = [x_{i,0} \; x_{i,1} \; \cdots \; x_{i,n}]$$

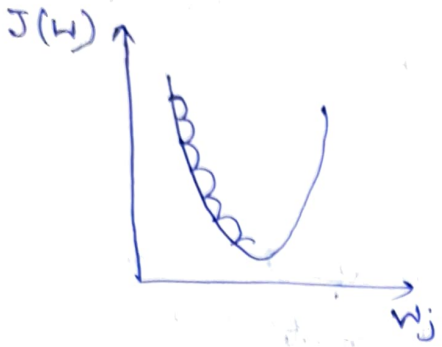
$$X = \begin{bmatrix} 1 & x_{1,1} & x_{1,2} & \cdots & x_{1,n} \\ 1 & x_{2,1} & x_{2,2} & \cdots & x_{2,n} \\ \vdots & \vdots & & & \vdots \\ 1 & x_{m,1} & x_{m,2} & \cdots & x_{m,n} \end{bmatrix}$$

$$\hat{y}_i = \sum_{j=0}^{n} w_j x_{i,j} \qquad x_{i,0} = 1$$

$$J(W) = \frac{1}{2m} \sum_{i=1}^{m} (y_i - \hat{y}_i)^2 \quad - \text{MSE / RSS}$$

Try to estimate $\hat{W}$ by minimizing $J(W)$

- It is a convex error function
- Geometrically, error surface is bowl shaped.




Simultaneously update
$j = 1, 2, 3, \ldots n$

$$w_j^{itr+1} = w_j^{itr} - \alpha \frac{\partial}{\partial w_j} J(W)$$

$\alpha$ → learning rate 0 to 1

$$\frac{\partial}{\partial w_j} J(W) = \frac{\partial}{\partial w_j}\left[\frac{1}{2m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2\right]$$

$$= \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)(-x_{i,j})$$

$$w_j^{itr+1} = w_j^{itr} - \alpha \frac{1}{m}\sum_{i=1}^{m}(\hat{y}_i - y_i)\, x_{i,j}$$

epoch: training with all samples.

## Algorithmic representation of GD

Step1: Randomly initialize the parameters $w_0, w_1, \ldots w_n$ and learning rate.

Step2: Calculate $\hat{y}_i$ for all training samples

Step3: Calculate $J^{itr}(w)$ for $itr^{th}$ epoch

Step4: Update all parameter $w_0, w_1, \ldots, w_n$

Step5: Repeat step 2-4 until

$$|J^{itr}(w) - J^{itr-1}(w)| > \rho$$

$\rho$: Small value.

## Variant of GD

### Batch gradient descent

→ All training samples are taken into consideration for single updating of all the parameters.

→ problem with huge training sample.

### Stochastic gradient descent

→ Update parameter by considering just one training sample at a time (repeat epoch by epoch till convergence.

$$w_j^{itr+1} = w^{itr} + \alpha(y_i - \hat{y}_i) x_{ij}$$

### Mini-batch GD

- Use small batch of training samples for updation.

- Difference between all ~~training~~ variant of GD. (Self Study)

| Parameter | Hyperparameter |
|---|---|
| Model parameter that can be estimated by fitting the given data | → Values set before the model start training |
| $w_0, w_1, \ldots w_n$ | → Can not be learned by fitting the model. |
| | $\alpha$, $\rho$, number of epochs. |